3.8.8 (a) Supply details to proof of (3.16)

$$\Gamma_1(N)\begin{pmatrix}1&0\\0&p\end{pmatrix}\Gamma_1(N) = \left\{\gamma\in M_2(\mathbb{Z}) : \gamma \equiv \begin{pmatrix}1&*\\0&p\end{pmatrix}(N),\ \det\gamma = p\right\}.$$

Pf: It is clear that LHS $\subset$ RHS. ~~$L_0 = e_1 \oplus e_2$~~.

RLHS $\supset$ RHS: Let $L = \mathbb{Z}^2$, $L_0 = L_0(N) = \left\{\begin{pmatrix}x\\y\end{pmatrix}\in L : y\equiv 0\ (N)\right\}$

Then $M_2(\mathbb{Z}) \curvearrowright L$ $\begin{pmatrix}a&b\\c&d\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}ax+by\\cx+dy\end{pmatrix}$.

$\forall \gamma\in M_2(\mathbb{Z})$ and $\gamma \equiv \begin{pmatrix}1&*\\0&p\end{pmatrix}(N)$, $\det\gamma = p$ $\quad \gamma L_0 \subset L_0$.

$$[L:\gamma L_0] = [L:L_0]\cdot[L_0:\gamma L_0] = N\cdot p.$$

There exists u, v a basis of L $\quad \det\begin{pmatrix}u\\v\end{pmatrix}(u,v) = 1$

and $\gamma L_0 = mu\mathbb{Z}\oplus nv\mathbb{Z}$.

and $0 < m \mid n$, $mn = N\cdot p$.

(Smith normal form is $\begin{pmatrix}d_1&&\\&\ddots&\\&&d_n\end{pmatrix}$ d_i, invariant ~~form~~ factor).

construction: if $\gamma L_0 = (a,b)\mathbb{Z}\oplus(0,c)\mathbb{Z}$

then $= (\gcd(a,b), 0)\mathbb{Z}\oplus(0,c)\mathbb{Z}$.

$L = u\mathbb{Z}\oplus v\mathbb{Z}$

$L_0 = u\mathbb{Z}\oplus pv\mathbb{Z}$

$\Downarrow$

$\gamma L_0 = u\mathbb{Z}\oplus Npv\mathbb{Z}$

Let $\gamma \equiv \begin{pmatrix}a_\gamma&*\\0&d_\gamma\end{pmatrix}(N)$ then $\begin{pmatrix}a_\gamma\\0\end{pmatrix} = \gamma e_1 \equiv \begin{pmatrix}0\\0\end{pmatrix}(m)$.

thus $\gcd(a_\gamma, N) = 1$ gives $\gcd(m, N) = 1$.

Since p prime, $\begin{cases} n = N\cdot p \\ m = 1\end{cases}$ so $\gamma L_0 = u\mathbb{Z}\oplus Npv\mathbb{Z}$.

Thus $L_0 = u\mathbb{Z}\oplus Nv\mathbb{Z}$, $\gamma L = u\mathbb{Z}\oplus pv\mathbb{Z}$.

Construct ~~thus~~ $\gamma_1 = (u\ v) \in \Gamma_0(N)$, $\gamma_2 = \left(\gamma_1\begin{pmatrix}1&0\\0&p\end{pmatrix}\right)^{-1}\gamma$

gives $\gamma = \gamma_1\begin{pmatrix}1&0\\0&p\end{pmatrix}\gamma_2$.

(3.8.d.(a)) Claim 1: $\gamma_2 \in \Gamma_0(N)$.

Claim 2: if $\gamma_1 \in \Gamma_1(N)$ then $\gamma_2 \in \Gamma_1(N)$.

Proof of claim 1: $\left.\begin{array}{l}\gamma e_1 \in \gamma L_0 \\ \gamma e_2 \in \gamma L_0\end{array}\right\} \Rightarrow \left\{\begin{array}{l} au + cpv \in u\mathbb{Z} \oplus Npv\mathbb{Z}. \\ bu + dpv \in u\mathbb{Z} \oplus pv\mathbb{Z}.\end{array}\right.$

$\Rightarrow a \in \mathbb{Z},\ c \in N\mathbb{Z},\ b \in \mathbb{Z},\ d \in p\mathbb{Z}$.

Thus $\gamma_2 \in \Gamma_0(N)$. which gives $\gamma_2 \in \Gamma_0(N)$.

Proof of claim 2: ~~Th~~ $a_\gamma \equiv 1\ (N) \Rightarrow \gamma e_1 \equiv e_1\ (N)$

$\Rightarrow$ ~~$au_1 \equiv 1\ (N)$~~ given $\gamma_2 = \begin{pmatrix} a & b \\ c & d\end{pmatrix}$

$au_1 \equiv 1\ (N)$.

$\gamma_1 \in \Gamma_1(N) \Leftrightarrow d_{\gamma_1\begin{pmatrix}1&0\\0&p\end{pmatrix}} = p. \Leftrightarrow d_{(\gamma_1\begin{pmatrix}1&0\\0&p\end{pmatrix})^{-1}} = \frac{1}{p}$.

$\Leftrightarrow d_{\gamma_2} = 1$.

Thus we proved RHS $\subset \Gamma_0(N)\begin{pmatrix}1&0\\0&p\end{pmatrix}\Gamma_0(N)$.

and if RHS $\subset \Gamma_1(N)\begin{pmatrix}1&0\\0&p\end{pmatrix}\Gamma_0(N)$ then we're done.

So now prove RHS $\subset \Gamma_1(N)\begin{pmatrix}1&0\\0&p\end{pmatrix}\Gamma_0(N)$.

$\Gamma_1(N) \backslash \Gamma_0(N) = \left\{ \begin{pmatrix} a & k \\ N & d\end{pmatrix} + \Gamma_0(N) \in SL_2(\mathbb{Z}) \right\}$. ~~$+\Gamma_0(N)$~~.

So suffices to show $\forall \begin{pmatrix} a & k \\ N & d\end{pmatrix}\ \exists \delta \in \Gamma_1(N)$

$\begin{pmatrix} a & k \\ N & d\end{pmatrix}\begin{pmatrix}1&0\\0&p\end{pmatrix}\Gamma_0(N) = \delta\begin{pmatrix}1&0\\0&p\end{pmatrix}\Gamma_0(N)$.

that is, $\exists \delta' \in \Gamma_1(N)\ \begin{pmatrix}1&0\\0&p\end{pmatrix}^{-1}\delta'\begin{pmatrix} a & k \\ N & d\end{pmatrix}\begin{pmatrix}1&0\\0&p\end{pmatrix} \in \Gamma_0(N)$.

If $p \mid N$, then $\delta' := \begin{pmatrix} dN+1 & -1 \\ -dN & 1\end{pmatrix}$ works. $c = -adN + N$.

If $p \nmid N$, then $\delta' := \begin{pmatrix} * & * \\ N & d'\end{pmatrix}$ $d' \equiv 1\ (N)$, $\equiv a\ (p)$ works.

$c = aN + Nd'$